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MILITARY SPECIFICATION

REEL, FUELING HOSE, AERIAL, REFUELING, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 SCOPE.- This specification covers the general requirements for a hose reel assembly suitable for use in a tanker aircraft and capable of refueling a receiver aircraft by the probe-drogue method of refueling.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-S-3136	Standard Test Fluids, Hydrocarbon
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-W-5086	Wire, Electrical, 600-Volt, Copper, Aircraft
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-5315	Packing, "O" Ring, Hydrocarbon Fuel Resistant
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design, Installation, and Data Requirements for

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MIL-G-5572	Gasoline, Aviation, Grades 80/87, 91/96, 100/130, 115/145
MIL-J-5624	Jet Fuel, Grades JP-3, JP-4, and JP-5
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-N-25027	Nut, Self-Locking, 250°F, 500°F, and 800°F
MIL-C-25162	Coupling, Reception, Flight Pressure Refueling, Type MA-2
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MS20995	Wire, Lock
MS29512	Gasket - Fuel Resistant Straight Thread Tube Fitting, Boss
MS29513	Packing "O" Ring Hydrocarbon Fuel Resistant
MS33586	Metals, Definition of Dissimilar
MS33588	Nuts and Plate Nuts, Self-Locking, Aircraft Design and Usage Limitations of

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

438	Age Controls for Synthetic Rubber Parts

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provision for preproduction testing.

3.2 COMPONENTS.- The components of the reel shall be as specified in the model specification.

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 MATERIALS.- Materials and processes used by the manufacturer of the reel shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. The use of contractors' specifications shall not constitute a waiver of Government inspection. All materials used in the reel shall be sufficiently resistant to fluids conforming to MIL-J-5624, MIL-G-5572, and MIL-S-3136 (type I and III), of aromatic content from 0 to 30 percent, to assure satisfactory operation as defined herein.

3.4.1 METALS.- Metals shall be of the corrosion-resistant type or be suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. The use of any protective coating that will crack or scale with age or extremes of climatic and environmental conditions shall be avoided. Metals shall be inspected in accordance with FED-STD-151. The use of magnesium or any alloy thereof is prohibited.

3.4.1.1 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals shall be as defined in MS33586.

3.4.2 CASTINGS.- Castings shall be high grade, clean, sound, and free from blow-holes, porosity, cracks, and any other defects.

3.4.3 FUNGUS-PROOF.- Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity, however, if they are to be used in a hermetically sealed inclosure, fungicidal treatment is not necessary.

3.4.4 SYNTHETIC RUBBER PARTS

3.4.4.1 AGE CONTROLS.- Age controls for synthetic-rubber parts shall comply with the requirements of ANA Bulletin 438.

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3.4.4.2 SERVICEABILITY.- All synthetic-rubber parts shall be readily replaceable with a minimum replacement of attaching parts.

3.5 DESIGN.- The reel shall be so designed that it may be mounted in bomber or cargo aircraft, or in an exterior store of fighter aircraft, using the tanker aircraft for the fuel and power supply. The reel shall be capable of refueling a probe-drogue type receiver aircraft.

3.5.1 STRUCTURAL AND MECHANICAL.- The reel shall be designed to withstand 115 percent of the loads encountered during refueling and aircraft operation without incurring permanent set that will adversely affect the mechanical operation of the unit. In addition, the hose reel shall be designed to withstand 150 percent of the loads without structural failure. The loads shall be specified in the model specification.

3.5.2 "O" RING PACKINGS.- If used, "O" ring packings shall conform dimensionally to MS29512 or MS29513 as applicable, and physically to MIL-P-5315.

3.5.3 LIFE AND MAINTENANCE.- The reel shall not require lubrication, replacement, or maintenance during 500 hours of operation under the conditions specified herein. The hydraulic fluid used in the drive assembly may be replaced at more frequent intervals but shall not require replacement during flight of the aircraft. Any lubricated part shall be a sealed replaceable assembly which has been lubricated in accordance with MIL-L-6880 or with a lubricant acceptable to the procuring activity.

3.5.4 ELECTRICAL REQUIREMENTS.- The electrical requirements shall be kept to a minimum and shall conform to MIL-STD-704. They shall also be specified in the model specification.

3.5.4.1 ELECTRICAL CONNECTIONS.- All external electrical connections shall be terminal connections conforming to MIL-T-7928 or standard connectors conforming to MIL-C-5015.

3.5.4.2 WIRE.- Internal wire shall be in accordance with MIL-W-5086.

3.5.5 DRIVE UNIT.- The reel shall be driven hydraulically. The hydraulic power requirements of the reel shall conform to the performance requirements and shall be specified in the model specification.

3.5.6 CONTROL.- The reel shall be operated automatically except for a trail signal at the start of the operation and a rewind signal upon conclusion of the operation. The reel shall not require adjustment in flight or change in drogue throughout the altitude-speed envelope specified in the model specification.

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3.5.7 REEL OPERATION.- The reel shall have sufficient response and winding rate so that excessive slack will not occur in the hose upon engagement by a receiver aircraft which has a rate of closure speed relative to the tanker aircraft of 10 feet per second. After contact, the reel shall automatically maintain 80 ±5 percent of the hose tension that existed in the free-trail condition. Response lag shall be sufficient to insure engagement of probe and drogue at closure speeds of 1 to 10 feet per second. Friction and hysteresis in the reel cycle shall not cause inadvertent disconnects when the receiver aircraft is maneuvering within the refueling envelope with a reception coupling disconnect force of 400 ±25 pounds.

3.5.8 RECEPTION COUPLING.- The reel shall be designed for utilization of a standard MA-2 reception coupling in accordance with MIL-C-25162.

3.5.9 SERVING GEAR.- A serving gear shall be incorporated on the reel to insure proper positioning of the hose coils on the drum.

3.5.10 EXTENSION OR RETRACTION SPEED.- The time for extension or retraction of the hose shall not be more than 15 seconds for each 60 feet of hose.

3.5.11 HOSE JETTISON.- Means shall be provided for quick release of the hose from the aircraft when conditions prevent rewinding of the hose on the hose reel. The hose jettisoning unit shall be designed to seal off the end of the hose retained on the hose reel.

3.5.12 SIGNAL LIGHTS.- The hose reel assembly shall have provisions for connecting signal lights, as an aid to the receiver aircraft pilot. A yellow light shall be a ready-for-contact signal and a green light shall be a flow-of-fuel signal.

3.5.13 HYDRAULIC SYSTEMS.- The hydraulic operating pressure of the hose reel fuel system shall be 3000 psi and shall otherwise conform to the requirements of MIL-H-5440.

3.5.14 ALTITUDE.- The reel assembly shall be designed to operate satisfactorily at any altitude from sea level to the maximum altitude specified in the model specification.

3.5.15 AIRSPEED.- When in a trail position, the reel assembly shall operate satisfactorily over the indicated airspeed range specified in the model specification.

3.5.16 FLOW.- The rated flow shall be specified in the model specification.

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3.6 CONSTRUCTION.- The reel shall be built to withstand the loads, strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The reel shall be so constructed that adjustment and repair can be made easily by the personnel of operating units and overhaul bases.

3.7 PERFORMANCE.- The reel shall satisfy the following performance requirements:

3.7.1 ENDURANCE.- The reel shall withstand cycling for a minimum of 500 cycles. There shall be no failure because of permanent set, deformation, or undue wear.

3.7.2 PRESSURE LOSS.- The pressure drop between the reel inlet and the reception coupling shall not exceed 70 psi at rated flow.

3.7.3 PROOF PRESSURE.- The reel shall withstand a hydrostatic proof pressure of 360 psi for 1 minute. There shall be no evidence of leakage, distortion, or other injury to any part of the system.

3.7.4 ENVIRONMENTAL RESISTANCE.- The reel shall conform to all requirements herein after exposure to the following conditions:

a. Vibration forces incident to service use. After testing, there shall be no failure because of permanent set, deformation, or undue wear.

b. Relative humidity up to 100 percent. After testing, there shall be no evidence of deterioration.

c. Exposure to salt-laden atmosphere. After exposure, there shall be no evidence of corrosion. Any corrosion shall be cause for rejection.

3.7.5 COMPONENTS, ENDURANCE.- All components of the reel shall withstand an endurance test. After testing, there shall be no leakage or malfunction of any of the components.

3.7.6 FUNCTIONAL.- After testing in accordance with the model specification, there shall be no leakage or malfunction during any one of the operating cycles. Leakage or malfunction shall be cause for rejection.

3.8 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

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3.9 THREADED CONNECTIONS

3.9.1 PIPE THREADS.- Pipe threads shall not be used on fuel system components except for permanent closures; in which case, the pipe threads shall conform to MIL-P-7105.

3.9.2 SCREW THREADS.- Machine screw threads shall conform to MIL-S-7742.

3.9.3 LOCKING OF PARTS.- All threaded parts shall be positively locked by safety wiring, self-locking nuts in conformance to MIL-N-25027, cotter pins, or other approved methods. Safety wire shall have a minimum diameter of 0.032 inch and shall conform to MS20995. Self-locking nuts shall be in accordance with MS33588. Staking and the use of lockwashers is not permitted.

3.10 DRAWINGS AND DATA.- The contractor shall furnish the following preliminary drawings to the procuring activity with submission of the model specification.

a. Reel assembly drawing.

b. Installation drawing.

3.10.1 The manufacturer's drawings shall consist of a cutaway section showing all parts in their normal assembled position; the drawings shall specify part numbers of all parts and subassemblies. The following data shall be furnished on or together with the assembly drawing:

a. Mounting dimensions

b. Overall dimensions

c. Location, size, and type of plumbing and electrical connections

d. Materials or construction, treatment, and finish.

3.11 WEIGHT.- The weight of the reel assembly shall be held to a minimum in accordance with general aircraft practice, consistent with reliability, and shall be specified in the model specification.

3.12 FINISH

3.12.1 ANODIZING.- All aluminum-alloy parts shall be anodized in accordance with MIL-A-8625, or adequately treated in some other acceptable manner for corrosion prevention.

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3.12.2 PROTECTIVE TREATMENT.- Steel parts, other than corrosion-resisting steel, not in moving contact, shall be cadmium plated in accordance with QQ-P-416, type II, class B, or be adequately treated in some other acceptable manner for corrosion prevention. Where galling of cadmium plating would cause malfunctioning, all wearing surfaces shall be chromium plated in accordance with QQ-C-320 or be adequately treated in some other acceptable manner for corrosion prevention. Brass, bronze, and copper-alloy parts shall be cadmium plated or be adequately treated in some other acceptable manner for corrosion prevention. Stainless-steel parts shall be passivated.

3.12.3 PAINT.- Paint shall be used for color code identification or as specified by the procuring activity.

3.13 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13.1 SYNTHETIC RUBBER PARTS.- Equipment and assemblies containing synthetic rubber parts used in aircraft fuel systems shall be marked in accordance with ANA Bulletin 438.

3.14 WORKMANSHIP.- All details of workmanship shall be of sufficiently high grade to insure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of the reel shall be classified as follows:

a. Preproduction tests

b. Acceptance tests

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4.3 TEST CONDITIONS

4.3.1 ATMOSPHERIC CONDITIONS.- Unless otherwise specified, all tests required by this specification shall be conducted at a room and fuel temperature between 60° and 90°F, at a barometric pressure of 28 to 32 inches of mercury, and at a relative humidity of 80 percent or less. Where tests are substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.3.2 CLEANING.- Before testing the reel, all oil and grease or other corrosion-resistant compounds shall be removed from the interior and exterior parts of the unit.

4.3.3 FUEL.- Unless otherwise specified herein, the fuel used during testing shall conform to MIL-J-5624, grade JP-4.

4.4 PREPRODUCTION TESTING

4.4.1 TEST SAMPLE.- The test sample shall consist of two reels representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 TEST REPORT, AND TEST SAMPLE.- When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report.- Three copies of a test report in accordance with MIL-T-9107.

b. Test sample.- The samples that were tested.

4.4.3 PREPRODUCTION TESTS.- The preproduction tests shall consist of all tests described under 4.6, TEST METHODS, conducted in the order listed unless otherwise specified in the test procedure:

REEL No. 1:

a.	Examination of product	4.6.1
b.	Structural	4.6.2
c.	Vibration	4.6.5
d.	Humidity	4.6.8
e.	Salt spray	4.6.9
f.	Component endurance	4.6.10
g.	Disassembly and inspection	4.6.13

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REEL No. 2:

a.	Examination of product	4.6.1
b.	Reel operation	4.6.3
c.	Endurance	4.6.4
d.	Pressure loss	4.6.6
e.	Fuel resistance	4.6.11
f.	Disassembly and inspection	4.6.12

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of the individual tests.

4.5.1 INDIVIDUAL TESTS.- Each reel shall be subjected to the following tests as described in 4.6:

a.	Examination of product	4.6.1
b.	Proof pressure	4.6.7
c.	Functional	4.6.12

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- Each reel shall be subjected to inspection for dimensional requirements in accordance with applicable drawings and shall be examined to determine conformance to the requirements of this specification with respect to material, weight, marking, use of the specified connections, finish, and workmanship.

4.6.2 STRUCTURAL.- The reel assembly shall be subjected to a load equal to

$$1.15\ \sqrt{(D + 1.5\ P)^2 + (W - L)^2},$$ where

D = Aerodynamic drag of hose and drogue at sea level and rated airspeed at normal trail.
P = Pull-off load due to maximum toggle setting.
W = Weight of hose with fuel and reception coupling.
L = Aerodynamic lift of hose at normal trail.

This load shall be as stated in the model specification. These loads shall be applied from the circumferential extremities of a 40° cone, centered about a position representative of the normal lay of the hose when extended during flight. A minimum of 5 positions shall be selected for the test, one of which shall be on the center line of the cone.

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4.6.3 REEL OPERATION.- It shall be demonstrated that the reel has a response and rewinding rate that will prevent excessive slack in the hose upon engagement with a receiver aircraft when closure speed relative to the tanker is 10 feet per second and, without further adjustment, permit engagement of the probe with the drogue when the closure speed is as low as 1 foot per second. It shall be further demonstrated that after contact the reel is capable of rewinding to 80 ±5 percent of the free-trail tension and can maintain that tension throughout the refuel operation until disconnect. For the purpose of this demonstration, it shall be assumed that the receiver aircraft starts to decelerate 1 second after contact. Deceleration shall be assumed to be at a constant rate, a rate that will equalize the speed of the receiver aircraft and the tanker aircraft within 6 seconds after contact. It shall also be demonstrated that friction and hysteresis in the reel cycle shall not cause inadvertent disconnects when the reel is operating in the refueling envelope with a reception coupling disconnect force of 400 ±25 pounds.

4.6.4 ENDURANCE.- The reel shall be cycled a minimum of 500 times. The tensile loads to be applied to the hose during the cycles shall be as specified in the model specification.

4.6.5 VIBRATION.- A vibration test shall be conducted in accordance with MIL-E-5272, procedure I.

4.6.6 PRESSURE LOSS.- A pressure loss test shall be conducted from 0 to 110 percent of the rated flow with a reel inlet pressure of 110 psi. The pressure drop from the inlet of the reel to the inlet of the reception coupling (drogue) shall not exceed 70 psi at rated flow.

4.6.7 PROOF PRESSURE.- The fuel system of the reel assembly shall be subjected to a hydrostatic proof pressure. A proof pressure of 360 psi shall be maintained for a minimum of 1 minute.

4.6.8 HUMIDITY.- A humidity test shall be conducted in accordance with MIL-E-5272 of procedure I.

4.6.9 SALT SPRAY.- The reel assembly shall be subjected to a salt spray test in accordance with MIL-E-5272 for a period of 50 hours.

4.6.10 COMPONENTS ENDURANCE.- All fuel system components, as agreed upon between the contractor and the procuring activity, having a definite operating cycle and which normally operate less than 10 times per flight

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shall be subjected to a minimum of 10,000 cycles; all others shall be subjected to a minimum of 100,000 cycles. The parts shall be operated at a working pressure of 120 psi during the wet cycles. This test shall consist of the following steps:

a. Wet endurance (room temperature): 50 percent of total cycles.

b. Dry endurance (room temperature): 20 percent of total cycles.

c. Wet endurance (fuel and ambient temperature at -65°F): 15 percent of total cycles.

d. Wet endurance (fuel and ambient temperatures at +135°F): 15 percent of total cycles.

4.6.11 FUEL RESISTANCE.- All components of the reel assembly that come in contact with fuel shall be tested for fuel resistance in accordance with table I.

4.6.12 FUNCTIONAL.- Each reel assembly shall be functionally tested in accordance with the model specification.

4.6.13 DISASSEMBLY AND INSPECTION.- The reel shall be disassembled and inspected. There shall be no evidence of deterioration, corrosion, or undue wear that might affect performance of the reel.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVEL A.- Reels shall be preserved and packaged in accordance with MIL-P-116, method 1A. The unit package quanity shall be one each.

5.1.2 LEVEL C.- Reels shall be preserved and packaged in accordance with the manufacturer's standard commercial practice.

5.2 PACKING

5.2.1 LEVEL A.- Reels preserved and packaged to meet 5.1.1 shall be packed in exterior overseas type shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or PPP-B-636, class 2.

5.2.2 LEVEL B.- Reels preserved and packaged to meet 5.1.1 shall be packed in exterior domestic type shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or PPP-B-636, class 1.

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TABLE I

Fuel Resistance and Extreme Temperature Test Schedule
For Fuel System Components Incorporating Electrical Equipment

Test	Fuel Resistance				
Period 1/	Phase I Soak	Phase I Dry	Phase II Soak	Phase II Dry	Low Temperature
Component configuration	2/	Drained, normal condition as expected under service conditions, ports open	2/	Drained, normal condition as expected under service conditions, ports open	Mounted as expected under normal service conditions 2/
Test fluid	MIL-S-3136, Type III	None	MIL-S-3136, Type III	None	MIL-S-3136, Type I
Period duration	96 hours (4 days)	24 hours	18 hours	30 hours	18 hours
Ambient and test fluid temperature	158° ±2°F or the normal operating temperature of the system in which the component is used, whichever is higher	158° ±2°F or the normal operating temperature of the system in which the component is used, whichever is higher	158° ±2°F or the normal operating temperature of the system in which the component is used, whichever is higher	158° ±2°F or the normal operating temperature of the system in which the component is used, whichever is higher	Immerse the component in MIL-S-3136, type I fluid and lower the fluid temperature to -67° ±2°F, then temperature at -67° ±2°F for a minimum of 18 hours

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TABLE I (Continued)

Fuel Resistance and Extreme Temperature Test Schedule
For Fuel System Components Incorporating Electrical Equipment

Test	Fuel Resistance				
Period 1/	Phase I Soak	Phase I Dry	Phase II Soak	Phase II Dry	Low Temperature
Operation or test during period	For intermittent duty components, actuate component 4 cycles/day. For continuous duty components and all pumps, run for 60 min/day	None	For intermittent duty components, actuate component 4 cycles. For continuous duty components and all pumps, run for 60 min	None	None
Operation or test immediately after period	Conduct leakage test using MIL-S-3136, type III fluid	(a) Conduct functional test (b) Actuate for 5 times or for 1 min, as applicable. (c) Conduct leakage test using MIL-S-3136, type I fluid	Conduct leakage test using MIL-S-3136, type III fluid	(a) Conduct functional test (b) Actuate for 5 times or for 1 min, as applicable. (c) Conduct leakage test using MIL-S-3136, type I fluid	With temperature no higher than -65°F, conduct functional and leakage tests using MIL-S-3136, type I fluid

1/ Each period shall follow immediately after the preceding one in the order noted.
2/ During periods of soaking in the test fluid, the component shall be maintained in such manner as to insure complete contact of all nonmetallic parts with the fluid as expected under normal service conditions.

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5.2.3 LEVEL C.- Packages that require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers and closures shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 MARKING.- Interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 AGE CONTROL MARKING.- Interior and exterior containers shall be marked in accordance with ANA Bulletin 438.

5.4 SHIPMENT MARKING NOMENCLATURE.- The shipment marking nomenclature shall be Reel, Fueling Hose, Aerial, Refueling.

6. NOTES

6.1 INTENDED USE.- The reel covered by this specification is intended for in-flight pressure refueling of receiver aircraft, using hydrocarbon fuel only.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Number of preproduction samples, point of testing, and delivery of preproduction test report (see 4.4.2).

c. Selection of applicable levels of preservation and packaging, and packing.

6.3 DEFINITIONS

6.3.1 CYCLE.- For the purposes of this specification, a cycle is defined as a start, trail, pull-off load, rewind, and stop, under the loads specified.

6.3.2 RECEPTION COUPLING.- The reception coupling comprises a thin-walled cylinder with valve and toggle assemblies, attached to the end of the refueling hose. Its primary function is to receive the probe of the receiver aircraft.

6.3.2.1 PULL-OFF LOAD MECHANISM.- The pull-off load mechanism is an integral part of the reception coupling and provides the necessary force required to keep the nozzle firmly engaged in the coupling. The assemblies are so designed that the force they apply can be varied.

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6.3.3 DROGUE.- The drogue, fabricated from light alloy sheet, is in the form of a 60-degree truncated cone. It is attached rigidly to the reception coupling by a flange and serves the function of stabilizing the hose and coupling when in the trail position, and provides a guide for the receiver aircraft probe.

6.3.4 PROBE.- The probe is the valve mechanism that is attached rigidly to the receiver aircraft. This valve is inserted into the reception coupling trailing from the tanker aircraft during the refueling operation.

6.3.5 EXCESS SLACK.- Excess slack is slack that will produce accelerated wear or damage to any component of the reel assembly or mating equipment.

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APPENDIX

10. SCOPE

10.1 This appendix establishes the form to be used by manufacturers in the preparation of a fueling hose reel model specification.

20. APPLICABLE DOCUMENTS

20.1 The following specification, of the issue in effect on date of invitation for proposals, forms a part of this appendix to the extent specified herein:

SPECIFICATIONS

Military

MIL-R-9126 Reel, Fueling Hose, Aerial Refueling, General Specification for

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. APPLICATION

30.1 A complete fueling hose reel model specification, in accordance with the outline and instructions for preparation as specified herein, shall be prepared for each specific model. New reel model designations will be assigned by the Government. When a new model designation is assigned, a new reel model specification shall be prepared with a new specification number assigned thereto by the unit manufacturer. It is desired that no new reel model designation be implemented by appendixes, or variants to an existing model specification and, in no case, shall amendments be used for this purpose. No change to a reel model specification shall be submitted to the procuring activity by means of amendments or revision pages prior to the time when the reel model specification is approved and becomes a part of a contract. Revisions by amendment form to an approved reel model specification, which has been released and is forming a part of a contract, will be acceptable to the procuring activity. Each amendment shall be approved and shall include and supersede the previous amendment.

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30.2 The headings and numbering of sections and paragraphs herein correspond to those of the basic specification for the specific data needed only in the model specification. Omission of reference in the model specification to a particular requirement of the basic specification shall be interpreted as compliance therewith. When departures are necessary from the requirements of this appendix and from the basic specification, the details of such departures shall be stated as specific requirements and shall bear the same section and paragraph heading and numbering as in the basic specification.

30.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the reel manufacturer for insertion of proper information and data related thereto in connection with the preparation of the model specification. Parenthetical statements shall not be copied verbatim in the model specification.

30.4 The specification number shall be the number assigned by the reel manufacturer. When revisions are made, they shall be designated by the use of a dash and a letter following the number, with a revision date therefor, which shall be shown on page 1 only. Only the specification number, and the revision suffix letter if applicable, shall be shown on subsequent pages.

30.5 For purposes of permitting preliminary evaluation of a proposed reel design or for release of approved reel performance characteristics in connection with an aircraft or aircraft equipment design competition, the reel manufacturer may submit a preliminary model specification to serve until superseded by a complete coordinated model specification which will be required for a production contract.

30.5.1 The preliminary model specification shall be prepared in accordance with the requirements stipulated herein.

40. MODEL SPECIFICATION

40.1 The form and description of the model specification follow:

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(Number and title.- The number and title shall be shown as follows:)

Spec No._____
(Date)_____
(a) Revised_____(Date)
(b) Revised_____(Date)

MODEL SPECIFICATION

REEL, FUELING HOSE, AERIAL REFUELING

(Insert service type and model designation if assigned)
(NAME OF CONTRACTOR)
(CONTRACTOR'S MODEL DESIGNATION)

1. SCOPE

1.1 SCOPE.- This specification covers the requirements for an aerial refueling hose reel.

1.2 CLASSIFICATION.- The fueling hose reel shall be of the following_____(Insert the contractor's model designation; also include a brief description of the reel and its operation.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for proposals, forms a part of this specification.

SPECIFICATIONS

Military

MIL-R-9126 Reel, Fueling Hose, Aerial Refueling, General Specification for

(List all additional Government publications not covered by this specification which are applicable to this reel.)

3. REQUIREMENTS

3.2 COMPONENTS.- The reel assembly consists of the following component parts: (State here in tabular form the component parts, such as reel, hose, drogue, reception coupling, and hydraulic controls.)

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3.5.1 STRUCTURAL AND MECHANICAL.- The reel and its components as noted shall be designed to withstand 115 percent of the following loads without incurring permanent set that will adversely affect the mechanical operation of the unit. The hose reel is also designed to withstand 150 percent of the following loads without structural failure: (List applicable loads encountered during flight and refueling operation, this should include "G" loads, drag loads and other applicable loads.)

3.5.4 ELECTRICAL REQUIREMENTS.- The electrical power requirements for the reel shall be as follows:

3.5.5 DRIVE UNIT.- The hydraulic power requirements of the reel shall be as follows:

3.5.6 CONTROL.- The following components of the reel assembly shall be controlled automatically. The operator shall control only the start of the trail operation and the start of the rewind operation.

3.5.14 ALTITUDE.- The reel assembly shall operate satisfactorily at any altitude from sea level to _____feet.

3.5.15 AIRSPEED.- The hose reel assembly shall operate satisfactorily over an indicated airspeed range from _____ to _____ knots.

3.5.16 FLOW.- The rated fuel flow for the reel assembly shall be _____gpm, at 70 psi pressure drop.

3.10 DRAWINGS AND DATA.- The following _____(insert manufacturer's corporate name) drawings and data form a part of this specification: (designate applicable drawings, as specified in MIL-R-9126.)

3.12 WEIGHT.- The weight of the hose reel assembly shall not exceed _____ pounds.

4. INSPECTION AND TEST PROCEDURES

4.1 The requirements for inspection and tests shall be as specified in MIL-R-9126.

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4.4.2 STRUCTURAL TEST.- The tensile load to be applied during the structural test shall be _____ pounds.

4.4.4 ENDURANCE TEST.- The tensile loads to be applied to the reel during the endurance test cycles shall be as follows:

4.4.12 FUNCTIONAL.- The functional test shall be performed on each reel assembly as follows: (State here the individual steps of the functional test such as trail, rewind, and automatic response.)

5. PREPARATION FOR DELIVERY

5.1 The requirements for preparation for delivery shall be as specified in MIL-R-9126.

6. NOTES

6.1 INTENDED USE.- The contractor should give a brief description of the intended application of the reel in the aircraft.

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NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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